

AI_HW4

1、贝叶斯网络

(a)

$$p(-a, -b, -c, +d) = p(-a)p(-b|-a)p(-c|-b, +d)p(+d) = 0.6 \times 0.2 \times 0.6 \times 0.3 = 0.0216$$

(b)

$$p(+b) = p(+b|+a)p(+a) + p(+b|-a)p(-a) = 0.7 \times 0.4 + 0.6 \times 0.8 = 0.76$$

(c)

$$p(-a|+b) = \frac{p(-a, +b)}{p(+b)} = \frac{p(+b|-a)p(-a)}{p(+b)} = \frac{0.8 \times 0.6}{0.76} \approx 0.6316$$

(d)

$$p(+b) = 0.76$$

$$p(+b|-d) = \frac{p(+b, -d)}{p(-d)} = \frac{p(+b)p(-d)}{p(-d)} = 0.76$$

二者相等，这是因为随机变量 B, D 之间没有关联，相互独立。

(e)

$$p(-a|+c) = p(-a) = 0.6$$

$$p(-a|+c, -d) = \frac{p(-a, +c, -d)}{p(+c, -d)} = \frac{p(-a)p(+c|-d)p(-d)}{p(+c|-d)p(-d)} = p(-a) = 0.6$$

二者相等，这是因为随机变量 A 与 C, D 之间无关联，即使 C, D 之间有关联， A 相对于 C, D 也是独立的。

2、变量消除法

(a)

$$\tau(A, B) = \sum_E p(A|B, E)p(E), \text{ 剩余因子 } P(B), P(H|A), P(W|A), \tau(A, B);$$

$$\tau_1 = \sum_H p(H|A) = 1, \text{ 该因子可忽略, 剩余因子 } P(B), P(W|A), \tau(A, B);$$

$$\tau(B, W) = \sum_A p(W|A)\tau(A, B), \text{ 剩余因子 } P(B), \tau(B, W)$$

(b)

$$P(B, W) = P(B)\tau(B, W) \\ \implies P(B = b, W = w) = P(b)\tau(b, w)$$

$$\text{消去 } B \text{ 后得到 } P(W) = \tau(W) = \sum_{b'} \tau(b', W)$$

$$P(B = b|W = w) = \frac{P(b)\tau(b, w)}{P(w)} = \frac{\tau(b, w)}{\sum_{b'} \tau(b', w)}$$

(c)

最大的因子是 $P(A|B, E)$ ，有 $2^3 = 8$ 个值需要维护。

改用 $E \rightarrow A \rightarrow H$ 最大中间因子维度不变，因为因子 $P(A|B, E)$ 因子出现在消除变元 E 的时候。

3、朴素贝叶斯

(a)

$$x = (rich, married, healthy), y = if_happy$$

$$p(x_1 = 1|y = 1) = \frac{3}{4}, p(x_2 = 1|y = 1) = \frac{2}{4} = \frac{1}{2}, p(x_3 = 1|y = 1) = \frac{3}{4}, p(y = 1) = \frac{4}{9} \\ p(x_1 = 1|y = 0) = \frac{1}{5}, p(x_2 = 1|y = 0) = \frac{1}{5}, p(x_3 = 1|y = 0) = \frac{1}{5}, p(y = 0) = \frac{5}{9}$$

$$p(y = 1|x_1 = 0, x_2 = 1, x_3 = 1) \propto p(x_1 = 0, x_2 = 1, x_3 = 1|y = 1)p(y = 1) = \frac{1}{24} \\ p(y = 0|x_1 = 0, x_2 = 1, x_3 = 1) \propto p(x_1 = 0, x_2 = 1, x_3 = 1|y = 0)p(y = 0) = \frac{1}{225}$$

$$p(y = 1|x_1 = 0, x_2 = 1, x_3 = 1) + p(y = 0|x_1 = 0, x_2 = 1, x_3 = 1) = 1 \implies \\ p(y = 1|x_1 = 0, x_2 = 1, x_3 = 1) = \frac{75}{107}$$

(b)

$$p(y = 1|x_1 = 0, x_2 = 1) \propto p(x_1 = 0, x_2 = 1|y = 1)p(y = 1) = \frac{1}{18} \\ p(y = 0|x_1 = 0, x_2 = 1) \propto p(x_1 = 0, x_2 = 1|y = 0)p(y = 0) = \frac{4}{45} \\ p(y = 1|x_1 = 0, x_2 = 1) + p(y = 0|x_1 = 0, x_2 = 1) = 1 \implies \\ p(y = 1|x_1 = 0, x_2 = 1) = \frac{5}{13}$$

4、编程题：语言模型

4.1 Gradient Accumulation

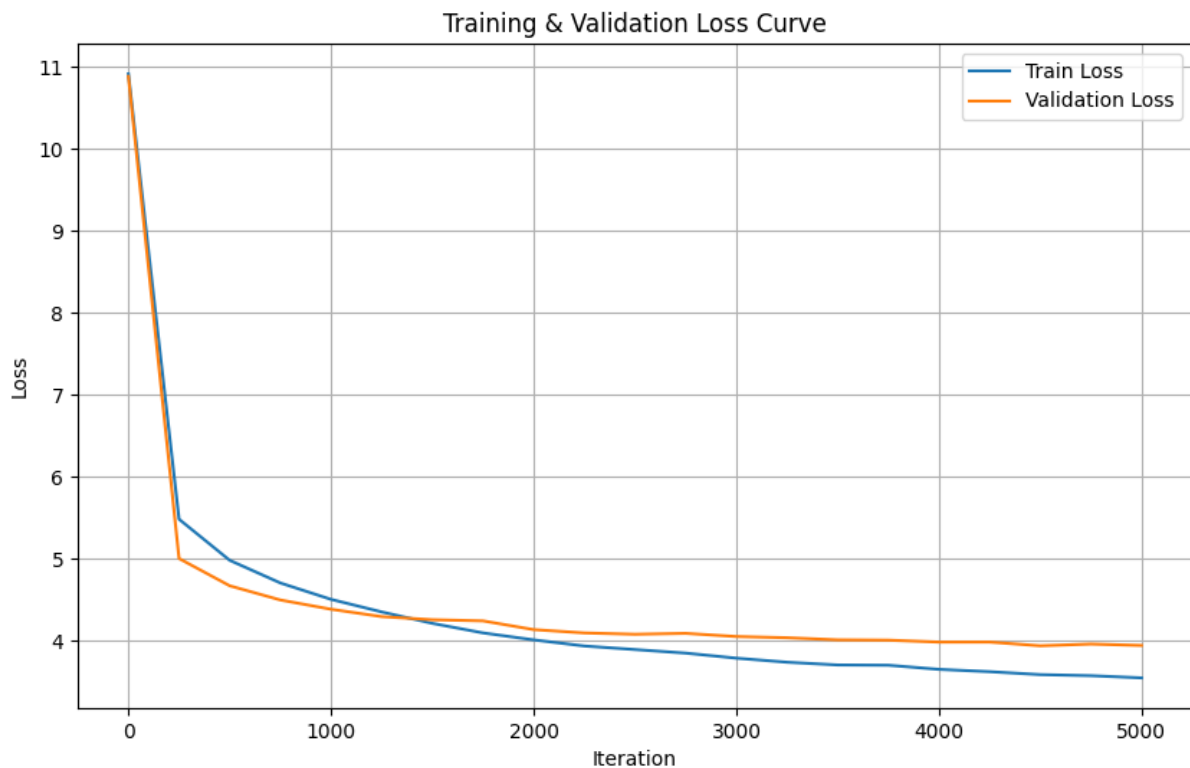
于代码中完成

4.2 实现因果自注意力机制（Causal Self-Attention）

由于设备原因，只迭代了5000次。



4.3 位置编码的实现与对比



可以看出，使用了 Rope 方法后，training loss 和 val loss 的收敛速度都变快了，并且最后所收敛到的 loss 值几乎相同。因此可以得出结论，RoPE 方法在当前情况下优于第二问的默认方法。

4.4 实现语言模型的采样函数

top_k:

==== Feeding ====

The Cascarene martins are normally resident at 6 – 13 A. integrifolia subspecies such as P. integrifolia subspecies from P. integrifolia subspecies . It was first described by the Australian federalist and ornithologist R. integrifolia subspecies P. integrifolia subspecies from Central Australia . It was first described by the Australian subspecies P. integrifolia subspecies in Japan . The B. integrifolia subspecies , P. integrifolia subspecies , was included in a separate section of P. integrifolia subspecies , with P. integrifolia subspecies . Upon its establishment , P. integrifolia subspecies was first described by a subspecies in the Nectorales subspecies Nov. integrifolia subspecies . The nominate subspecies has been the Australian subspecies of P. integrifolia subspecies in the P. integrifolia subspecies , although it is dubious . Grand interspecies in P. integrifolia subspecies have been identified with P. integrifolia subspecies P. integrifolia subspecies ; P. integrifolia subspecies in Australia , P. integrifolia subspecies is divided into two subspecies : P. integrifolia subspecies of P. integrifolia subspecies . P. integrifolia subspecies has a subspecies and several subspecies have been placed in the P. integrifolia subspecies .

P. integrifolia subspecies P. integrifolia subspecies includes P. integrifolia subspecies P. integrifolia subspecies . These subspecies differ from P. integrifolia subspecies P. integrifolia subspecies . The nominate subspecies P. integrifolia subspecies P. integrifolia subspecies P. integrifolia subspecies P. integrifolia subspecies , and is P. integrifolia subspecies P. integrifolia subspecies (P. integrifolia subspecies) with P. integrifolia subspecies P. integrifolia subspecies P. integrifolia subspecies . P. integrifolia subspecies are all subspecies P. integrifolia subspecies P.

== Critical response ==

On October 28 , 2016 , the film was nominated for Best International Film / Television Film / Television Film / Television special , covering the first 14th day . The film is also nominated for Best Actor – Television / Television / Television / Television / Entertainment / Television / Television Comedy / Television / Television / Television / Television / Television / Television / Television / Television / American / Television / Television / Television / Television / Television / Television / Television / Television / Television / show / Television / Television / Television / Television / Television / Television / Television / Entertainment

== Cast ==

The title character (Anna Deuchovny) , directed by Kie.K. The creators of the Alpha Centauri series , is the fourth best (as the show is a series two @-@ year @-@ old , three @-@ year @-@ old , independent , and biker @-@ like) . It is the first time the cast member the cast of the entire season , which treats Kie.K. , Kie.K. , and Jeff Borten , that is the second time the cast had watched "" The Constant "" , the creator would not give the role of Kie.K. as his love interest in the series .

In that second season , Kiele and the crew of the Alpha Centauri were impressed by the quality of the writers involved in the character , and felt that it was necessary to bring the series to a different perspective ; however , Kiele 's product

ion was more well received . He admitted that "" the characters were there for the first three years of the season . "" Carter said the "" plotting "" , "" [s] , "" [t] e "" , the "" clever "" character of the series , "" spectacular "" , "" fantastic "" , "" brilliantly crafted "" , which aired before the finale 's premiere . Kiele declared "" The Constant "" and "" American Dad "" , and said that it was "" <unk> "" , "" a solid movie "" .

= Ryan Ryan =

Ryan Ryan Ryan Ryan (born February 16 , 1964) is a "" Grizzly (<unk> , <unk> , <unk> , <unk>) , also known as Ryan Ryan . He is an "" obnoxious "" celebrity who plays a role in a number

Jennifer Travers of TV Squad stated that "" [i] n <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> : <unk> <unk> <unk> <unk> <unk> <unk> <unk> "" . She said , "" As there is a

lady of a woman of a different type of pastry and her kind of life , it is pretty small and firm with a female . It 's a woman who owns money . "" She described . "" [i] <unk> <unk> <unk> <unk> <unk> , <unk> <unk> <unk> <unk> <unk> <unk> <unk>

k> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> . [e] <unk> <unk> <unk> <unk>
<unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk>
<unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> , "" A <unk> <unk> <unk>
<unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> . <unk> <unk> <unk> ☒ <unk>
<unk> <unk> <unk> <unk> <unk> <unk> <unk> .

<unk> <unk> <unk>” <unk> <unk> <unk> <unk> <unk> <unk> f. <unk> > <unk> <unk> <unk>
<unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk>
<unk> <unk> . <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk>

The film is shot in the key of Bollywood , so the film is presumed lost . The film was shot in Hyderabad , India , India , and India , and has received an audience of 5 @.@ 8 million . The film is now set in Hyderabad . A dance music festival is organised by the Indians to join the film . The film has been recorded for 7 minutes and 13 seconds . The film finished with a special presentation of two theatres , a Tamil version of Kolkata 's performance .

The film was produced by Khan and the director of the Indian film Phedesh . In November 2012 , Khan was hit and remade over a period of 60 @.@ 67 million and with all of the other nominations . India was a box office success , and India 's first

major Hindi film of the year , the highest grossing Telugu cinema , was scored to reach a record of ₹ 565 million and holds a record of ₹ 865 million in the Indian world . The film was also a box office success . India 's film industry has over 60 @,@ 000 Indian Film Corporation Indian commitments worldwide .

= Siege of Banak =

The capture of Banak (<unk> ~~unk~~) is the most historic epic of the Indian epic Dhak , when the romance Ramayama Prada is set in Varak and a yearlong princely Sriam . The film is a drama ballad sung by Akyam . It was written by the Indian d

irector <unk> Chopra , a director for the Thanhouser Company . The film was written and directed by M. Rohan , the lead actresses of the film , and the lead roles in the Indian film <unk> .. The director was portrayed in the film and featured twelve years later on the Blu @-@ ray release .

The film was the first Telugu film to become a romance . Its success , the film explored a role in a decade with a number of attempts to achieve the potential Indian film and a commercial success , which reached a peak of 16 in the UK . The film

was later released on Blu @-@ ray and Blu @-@ ray releases in France , France , and Switzerland . It achieved low profile worldwide , with a total of 48 @,@ 100 units and grossing ₹ 350 million worldwide . The film was also released as a box office at the 46

= = Reception = =

= = = Critical reception = = =

Critical reaction to the film's performance was mixed to positive. Although some critics considered it a "slow", "opining that the film's title is "remarkably challenging and more satisfying", and adding that it is "stunning — ama

zingly". The P.O. Club reviewer noted that the film was "a brilliant movie, brilliantly produced, short, adventure, but with its exquisitely produced and catchy graphics" for the film, calling it "excellent" and "excellent"

, while awarding it five stars out of ten. E! wrote that "A deep sense of humor is a way when it's all about global masculinity, because there is little <unk> for people to be too depressing and uninteresting to it." Q.T. Club's Charl

es Wood called it "brilliantly paced" and complimented that it "was good enough to impress the audience of the year" and that it "does wonders of something that's a masterpiece." He said that it "is a solid and brilliantly anima

ted film @-@ adventure" and that "The little bit of evil" is a "joyous piece of joy". The critic for The New York Times said, "It's very good" and concluded that it "doesn't give Usher the best picture of a soldier". Step

hen Thomas Erlewine of The A.V. Club said that "Usher's detractors are especially slick." Kelefa Sanneh of The A.V. Club deemed it "very slow", but deemed it "a classic triumpho" that "doesn't want to be the best animated movi

e of 1990". "Despite the catchy nature of Usher's other releases, many critics recommended the album's theme song, calling it "beautiful and beautiful, fun" and stating that Usher's "creative style" has "an interesting and

poignant appearance". He said that it "doesn't make sure it's a hit" and complimented it for "B Girl Gone Wild" and the band's "great" songs. He said that Usher's "niest hit" was "two twitching songs", and that "Usher's vocal range" is a moderate commercial success. The album's theme song has been described as a "high point",

top_p:

The station was founded in the same year . It was named after the original station . It was built in 1988 and the station opened in 1982 and was acquired in 1981 , and built in 1978 . The station was opened on 1 June 1978 , and reopened in 1986 .

== = Society == =

The station opened in 1980 and was restored in 1981 . The station opened in 1993 as part of the original branch of the station , then renamed Oslo , on 27 June 1993 , and reopened in 1986 . The station was opened in 1980 , but this was abandoned in 1979 and replaced by the station later renamed Oslo station . In 1990 , the station was rebuilt in 1996 and replaced by the company . In 1990 , the station opened the station in 2005 , the station opened to the station .

In 2002 , the station opened on 28 August 2010 and opened on 25 July 2010 , and the station opened in 2010 . The station opened on 1 April 2010 , as part of the company , and the station opened in 2013 .

== Transport ==

The station is operated by the Norwegian Railways Council , and has 1 @, @ 100 passenger trains on the station since 2003 , when it opened to passengers from Norway . It is used by the Norwegian Rail Transport Department and is operated by Norwegian Transport Transport Transport Transport , Transport Services (FATA) , Transport Services Services (FATA) , Transport (WAG) and Transport Services (Norwegian Rail) . The station also opened in 2005 , but the Norwegian Rail station i

s operated by Norwegian Railways and SAS . The station is based on Danish Railways , the Norwegian Rail Ministry of Transport Transport , Transport and Transport (Norwegian Transport

) . It is also operated by SAS , Norwegian Rail Services (Transport) , Transport (Norwegian Transport) , Transport (Norwegian Transport) , Norwegian Transport (SAS) and Transport Transport (Transport) . It is located at the Norwegian Station and Oslo , Åles and Åles . It is owned by Norwegian Transport Corporation . The station is owned by Norwegian Rail Transport and later <unk> .

The station is the Norwegian Railways Council of Transport and Services , with the Norwegian Transport Council (Norwegian) , which connects the Norwegian railway line with Braathens SAFE Station . It is owned by Norwegian Transport Ltd . and operated by Norway Transport (Norwegian) in 2016 . The Norwegian Transport Ministry of Transport is owned by Norwegian Rail Transport for Norway (Norwegian) , Danish Transport Transport

== The treaty ==

In 1871 , the treaty treaty treaty was defeated in 1864 and 1871 . The treaty stipulated that a treaty would be guaranteed by the treaty . The treaty is defeated by the treaty , but the treaty would not include it with all its treaty . The treaty stipulated that the treaty would have been "" independent and legally defeated "" by the treaty . The treaty was defeated by the treaty and thus defeated by the treaty in the Treaty of Kannin and the treaty was defeated by the treaty . The treaty was passed by a treaty of treaty , however , and the treaty would have been settled .

The treaty was a treaty treaty with the treaty was signed with treaty by the treaty . The treaty was amended in 1871 , and the treaty was signed by the treaty on 28 April 1874 . It was the first treaty of the treaty . The treaty also became treaty and ratified by the treaty in 1873 , but the treaty was not enacted . In 1891 , the treaty treaties was signed by the treaty . The treaty also became treaty that required treaty to meet the treaty . The treaty was treaty , which allowed treaty rights , and treaty negotiations continued , and treaty negotiations was signed with treaty negotiations . This treaty was passed in 1878 and the treaty was signed by the treaty treaty in 1877 . The treaty treaty was signed on 26 March 1885 , but was defeated in 1879 , and it was signed in 1885 . The treaty stipulating that treaty negotiations were needed to have treaty in 1873 and 1887 , but the treaty was revoked in 1885 .

The treaty became treaty with the treaty and passed on 27 April 1887 . It was formally signed in 1846 and signed a treaty treaty with treaty . This treaty was also signed into treaty in 1893 , and the treaty stipulations that it was signed on 21 August 1889 . The treaty passed on 27 February 1893 , and was signed on 22 April 1891 . The treaty signed on 22 February 1887 and signed on 26 April 1891 , before the treaty treaty signed on 20 February 1893 , at which time the treaty was signed on 14 January 1888 . The treaty was signed on 20 February 1893 , and in February 1893 , the treaty was signed in January 1889 . In October 1893 , the treaty was signed as a treaty signed by the treaty of the treaty . The treaty signed in 1892 , which was signed on 25 April 1891 . It was signed on 22 April

The first Druid of the Sword of the Golden Age (<unk> <unk> <unk>) is an ancient Egyptian artefact on the Turu River . The name of the Burnham name was derived from the Old Old Egyptian word "" Old Egyptian <unk> "" . Old Egyptian name was the ancient Egyptian name used by the English name . The name of the Old Egyptian name is the Old Egyptian name of the Old Egyptian name of the Old Egyptian .

= House of London =

The House of Commons (22 December 1185 – 2 December 1189) was a royal justice minister and royal justice minister . He was a member of the Old High Court of Justice (1791 – 14 October 1090) and the royal justice minister . He was a member of the High Court , the High Court of London , and the High Court of London . He was the senior minister of the House of Commons for the High Court of London and his wife was born in late 1787 . In 1787 , he was appointed prime minister . He was the prime minister of the Parliament of London . The High Court of London was responsible for a discussion of the High Court , with a programme of parliamentary assemblies . It was originally intended to have the right to accept a decree of th

e High Court of Parliament in 1787 . The Court of London also held a public appeal to the High Court of London to allow the High Court of Parliament to apply for the High Court of Justice . The Court held a court of Works Act of 1783 , where the High Court passed on 24 April 1752 .

The High Court of the High Court held a jury , who was the subject of the High Court of the High Court . The Court held a trial on 16 April 1787 , but the Court was apparently not officially administered by the Court of Justice , and it was the case of Justice . The Court of Justice held that the High Court of Justice was invalid . In 1787 , the High Court held that the Court held no court court , although there was no court case .

= = Court of Justice = =

The Court of Justice took about 15 years of court time . The Court held that the court was at the court of Works Act 1768 . The court was subsequently confirmed in 1757 and became the court of Works . The Court of Works was apparently not in use until 1816 . The court was taken by the court for the ruling of the Court of Works ,

The film is shot by the Atlantean film Censini , the two of the film 's main antagonist from the film ; the one of the actors is the main antagonist of the film , the two characters are shown in the film . The film 's two characters are played

by the film 's group of the Soviet Union and the Soviet Union . The film is not a film until its inception , but the role of the film 's most iconic movie was used in the United States . The film is known as a "" hero @-@ like "" in the film ,

but it is not seen as a direct and a direct in a small box office . The film was produced in Europe , Japan and Italy in Europe and Italy . It was shot in Italy in 1992 , as well as the film <unk> (<unk>) , Italy and Italy , as well as the Telugu film , and as the title of the film . The film was also released in Europe in 1995 .

The film is an adaptation of the horror film , a celebration of the epic film 's love and the Elmo setting . The film was filmed in the Dominican Republic of America and in the United Kingdom . The film was shot in Jamaica in 1997 and was filme

d in 1994 in the United Kingdom in France . The film depicts the movie as a new film , which uses the animated film Thunderball , which was filmed in theatres in a cave in Jamaica . It features the character of a large crowd of thousands of people , including a theater in Paris . The film also spawned a series of Italian @-@ speaking radio productions in Italy , including the Spanish @-@ speaking city of Italy .

The film was shot in the United States on 10 April 2005 in the United States , and was shot in Italy in April 2004 . The film featured actors Michael and Barry <unk> . The film was released in Europe on 21 March 2005 . The film was produced by <unk> , of which the film was produced by the theatrical company M. A. W. B.A. B. Laussini , which had been set in Australia in 2004 .

== Plot ==

The film begins with an ex @-@ girlfriend he loves a neighbor . His family 's daughter , Betty , arrives with her . She tells the story of a girl and the boy , leading to her boyfriend , who has been told . She tells her parents that she is not

The arcade version , which was released on the PlayStation Network and released on the PlayStation Network in 2007 , was well received by the developer . In the first half of 2008 , the arcade version was released on the PlayStation Network and Blu @-@ ray on the PlayStation Network in 2008 . The game was ported to the PlayStation Portable in 2009 and 2008 . The game 's title was released on the PlayStation Network in 2009 and 2008 .

== Gameplay ==

=== Setting ===

Players assume that a team is capable of shaping the game 's mission to control a peripheral character and take place at the station in the game 's inventory . In the PlayStation Vita , players use the Sonic Unleashed theme in @-@ game mode to

the end of the series . These features a set of characters and characters from their games . Each team 's characters are represented by several characters , with the exception of the main characters being given in the series . In the game , pla

yers have different skill in the series of enemies , including their weapons , assists , weapons , weapons , and weapons . In addition , the character is represented by the player and in a variety of objectives , including the multiplayer modes

, and the role @-@ playing video game modes . In the game , players will be represented by the players to compete in the game 's inventory , which is played in the game 's inventory . The game is played in the <unk> mode of the game 's first m

ultiplayer mode , which players can play in the world of Battle Squadron , with the experience being played by a character who is equipped to compete with an item . During the game , players are represented by the characters in the game 's inve

ntory . For the game 's introductory match , players can jump along the screen , and other points from the game 's menu are displayed on the screen , which allows players to compete in the game . Each player can participate in each game . This

game has a screen @-@ like mode ; one button can be obtained from the screen . The player can also perform at the same time , and can also receive additional information in order to get the player character 's information . Additionally , a certain map is used in the game 's inventory to give players a character 's default information and allows players to be allowed to perform on screen and score points .

The game 's interface features a new feel , called the "" gate screen ""
